

HIGH PERFORMANCE COMPUTING

ASSIGNMENT-4

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Batch : 22

Assignment 1: Parallel Vector Addition Scenario Add two large vectors in parallel and compare serial and parallel execution time. Objective To understand basic OpenMP parallel loop execution. Tasks

1. Initialize two large vectors.
1. Implement vector addition using a serial loop.
2. Implement vector addition using a parallel loop.
3. Compare the execution time of serial and parallel implementations. Learning Outcomes ☐ Differentiate between serial and parallel program execution. ☐ Implement OpenMP-style parallel loops using Python.

☐ Measure and compare execution time of serial and parallel programs. ☐ Analyze the performance benefits of parallel computing.

```
import time
import numpy as np
from concurrent.futures import ThreadPoolExecutor

# Vector size
N = 10_000_000

# Initialize vectors
A = np.random.rand(N)
B = np.random.rand(N)
C = np.zeros(N)

# ----- SERIAL EXECUTION -----
start = time.time()
for i in range(N):
    C[i] = A[i] + B[i]
end = time.time()
print("Serial Time:", end - start)

# ----- PARALLEL EXECUTION -----def
def add_chunk(start, end):
    for i in range(start, end):
        C[i] = A[i] + B[i]
num_threads = 4
```

```

chunk_size = N // num_threads

start = time.time() with
ThreadPoolExecutor(max_workers=num_threads) as executor:
    for i in range(num_threads):
        executor.submit(
            add_chunk, i *
            chunk_size,
            N if i == num_threads - 1 else (i + 1) * chunk_size
        )
end = time.time()
print("Parallel Time:", end - start)

```

Serial Time: 8.984385251998901

Parallel Time: 3.983694076538086

Assignment 2: Static Scheduling in Parallel Loop Scenario Perform a vector computation using static scheduling. Objective To study static scheduling behavior in OpenMP-style parallel loops. Tasks

1. Initialize input arrays.
1. Perform vector computation using a parallel loop.
2. Measure execution time.
3. Observe work distribution among threads. Learning Outcomes
 - 1 Understand the concept of loop scheduling in OpenMP.
 - 2 Explain static scheduling and its work distribution mechanism.
 - 3 Implement static scheduling in parallel loops.
 - 4 Evaluate the effect of static scheduling on execution time.

```

import numpy as np import time from
concurrent.futures import ThreadPoolExecutor

```

```
N = 1_000_000
```

```
A = np.random.rand(N)
```

```
B = np.zeros(N)
```

```
num_threads = 4 chunk_size =
```

```
N // num_threads
```

```

def compute(start, end, tid): print(f"Thread {tid}
handles range {start} to {end}") for i in
range(start, end): B[i] = A[i] * A[i]

```

```
start_time = time.time() with
```

```
ThreadPoolExecutor(max_workers=num_threads) as executor:
```

```
    for tid in range(num_threads):
```

```
        s = tid * chunk_size
```

```
        e = N if tid == num_threads - 1 else (tid + 1) * chunk_size
```

```
        executor.submit(compute, s, e, tid)
```

```
end_time = time.time()
```

```
print("Execution Time:", end_time - start_time)
```

Thread 0 handles range 0 to 250000 Thread 1 handles range 250000 to 500000

Thread 2 handles range 500000 to 750000

Thread 3 handles range 750000 to 1000000

Execution Time: 0.4210777282714844

Assignment 3: Load Imbalance in Parallel Execution Scenario Analyze parallel loop performance when different iterations perform different amounts of work. Objective To understand load imbalance and its impact on scheduling. Tasks

1. Create an array of random numbers.
1. Perform light computation for some iterations.
2. Perform heavy computation for other iterations.
3. Measure and analyze execution time. Learning Outcomes ^[7] Identify load imbalance in parallel programs. ^[8] Understand how uneven workload affects performance. ^[9] Analyze the limitations of static scheduling for imbalanced workloads. ^[10] Relate load imbalance issues to scheduling strategies in OpenMP.

```
import random
import time
from concurrent.futures
import ThreadPoolExecutor
```

```
N = 100000
```

```
data = [random.randint(1, 100) for _ in range(N)]
```

```
def work(start, end, tid):
```

```
    for i in range(start, end):
```

```
        if data[i] < 50:
```

```
            sum(j*j for j in range(100))    # Light work
```

```
        else:
```

```
            sum(j*j for j in range(5000))    # heavy work
```

```
    print(f"Thread {tid} completed")
```

```
num_threads = 4 chunk =
```

```
N // num_threads
```

```
start = time.time() with
```

```
ThreadPoolExecutor(max_workers=num_threads) as executor:
```

```
    for tid in range(num_threads):
```

```
        executor.submit
```

```
            ( work, tid
```

```
              * chunk,
```

```
              N if tid == num_threads - 1 else (tid + 1) * chunk,
```

```
              tid
```

```
            ) end =
```

```
time.time()
```

```
print("Total
```

```
Time:", end -  
start)
```

```
Thread 3 completed  
Thread 2 completed  
Thread 0 completed  
Thread 1 completed  
Total Time: 22.646467208862305
```

Assignment 4: Parallel Reduction (Synchronization)

29-01-26 Thursday Scenario Compute the sum of all elements in an array using parallel execution. Objective To eliminate race conditions using reduction. Tasks

1. Initialize a large array.
1. Attempt to sum elements in parallel.
2. Use OpenMP-style reduction to avoid race conditions.
3. Display the final sum. Learning Outcomes

🔗 Identify race conditions in parallel programs. 🔗 Understand the need for synchronization in shared data access. 🔗 Apply reduction techniques to safely accumulate shared variables. 🔗 Implement correct and race-free parallel reductions.

```
import numpy as np  
from concurrent.futures import ThreadPoolExecutor  
  
arr = np.random.rand(1_000_000)  
num_threads = 4 chunk =  
len(arr) // num_threads  
  
def partial_sum(start, end):  
    return sum(arr[start:end])  
  
results = [] with ThreadPoolExecutor(max_workers=num_threads)  
as executor:  
    for i in range(num_threads):  
        s = i * chunk  
        e = len(arr) if i == num_threads - 1 else (i + 1) * chunk  
        results.append(executor.submit(partial_sum, s, e))  
  
final_sum = sum(r.result() for r in results)  
print("Final Sum:", final_sum)  
  
Final Sum: 499817.2489515851
```

Assignment 5: Barrier Synchronization (Two-Phase Computation) Scenario Perform a computation in two phases ensuring correct synchronization between phases. Objective To demonstrate barrier synchronization in parallel execution. Tasks

1. Perform computation in Phase 1.
1. Synchronize all threads.
2. Perform computation in Phase 2.
3. Display completion message. Learning Outcomes

🔗 Understand barrier synchronization in parallel computing. 🔗 Implement multi-phase parallel programs. 🔗 Ensure correct execution order using synchronization techniques. 🔗 Explain the role of barriers in OpenMP-style parallel execution.

```
from threading import Barrier, Thread
import time
barrier = Barrier(4)

def task(tid):
    print(f"Thread {tid} Phase 1 started")
    time.sleep(1)
    print(f"Thread {tid} Phase 1 finished")

    barrier.wait()      # 🔗 Barrier synchronization

    print(f"Thread {tid} Phase 2 started")
    time.sleep(1)
    print(f"Thread {tid} Phase 2 finished")

threads = []
for i in range(4):
    t = Thread(target=task, args=(i,))
    threads.append(t)
    t.start()

for t in threads:
    t.join()
print("All threads completed both phases")
```

```
Thread 0 Phase 1 started
Thread 1 Phase 1 started
Thread 2 Phase 1 started
Thread 3 Phase 1 started
Thread 0 Phase 1 finished
Thread 1 Phase 1 finished
Thread 2 Phase 1 finished
Thread 3 Phase 1 finished
Thread 3 Phase 2 started
Thread 0 Phase 2 started
Thread 1 Phase 2 started
Thread 2 Phase 2 started
Thread 0 Phase 2 finishedThread 3 Phase 2 finished
Thread 1 Phase 2 finished
Thread 2 Phase 2 finished
```

All threads completed both phases